

Extra — One Loop, Then Two

Topic: Loops · Counting · **Course:** Extra Worksheet · **Time:** about 30 minutes

Words to use: **loop, turn, row, inside, wave**

No code today. Only pictures. We start with **one** loop. At the end we put one loop inside another.

1 · A Loop Takes Turns

A loop does the same job again and again. Each time is one **turn**.

Here the job is: print one 🐙.

TURN 1

🐙

TURN 2

🐙

TURN 3

🐙

TURN 4

🐙

THE SCREEN

🐙

🐙

🐙

🐙

How many turns?

The loop takes 6 turns. How many 🐙 on the screen?

2 · Each Turn Takes One Monster 📦

Now the loop has a box of monsters. Each turn takes the **next** one.

TURN 1



first

TURN 2



next

TURN 3



next

TURN 4



last

THE SCREEN






Which monster comes on turn 3?

The box gets one more monster. How many turns now?

3 · How Long Is One Row?

One turn makes one row. The row can be short or long.

ROW LENGTH

 ← 1 monster

 ← 2 monsters

 ← 3 monsters

 ← 4 monsters

Draw a row of 5 .

Every row has 4 monsters. There are 4 rows. How many monsters?

4 · One Loop Makes One Wave 🌊

Four turns. Four rows. Every row has 4 monsters.

All of it together is **wave 1**.



How many rows in wave 1?

How many monsters in wave 1?

5 · A Loop Inside a Loop 🐧

One loop makes one wave. You want 4 waves.

So you put the loop **inside** another loop.

BIG LOOP · do the whole wave 4 times

SMALL LOOP · 4 turns · one row each

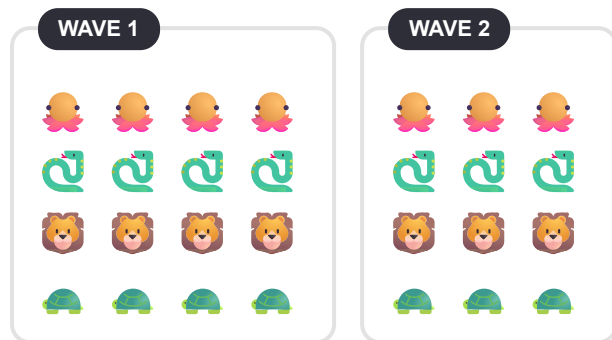


Which loop is inside?

The big loop takes 4 turns. The small loop takes 4 turns each time. How many rows in total?

6 · Every Wave Gets Smaller

Wave 1 rows have 4 monsters. Wave 2 rows have 3.



Wave 3 rows have _____ monsters. Wave 4 rows have _____ monsters.

The wave number goes up. What does the row do?

Draw wave 3 here. Keep 4 rows.

7 · Draw and Tell 🎨

Pick 3 monsters. Draw wave 1 with rows of 3. Then draw wave 2 with rows of 2.

Write the wave number next to each wave.



Then make a short video on your phone. Point at your picture and say:

1. How many rows are in wave 1.
2. What is different in wave 2.
3. Which loop is inside the other one.

Submit ✅

Send this worksheet + your photo or video to teacher on KakaoTalk.